

Government Hospital Resource Demand Forecasting System

Introduction

This project predicts hospital resource requirements using machine learning techniques. The system forecasts beds, ICU units, oxygen demand, doctors required, medicine units, and emergency level using historical hospital resource data.

Objectives

- Predict future hospital resource demand.
- Reduce resource shortages.
- Improve emergency preparedness.
- Provide real-time prediction dashboard.

Technologies Used

Frontend: HTML, CSS, JavaScript

Backend: FastAPI

Machine Learning: Scikit-learn Random Forest

Deployment: Render + Vercel

Programming Language: Python

Dataset Information

The dataset contains 5000 rows and 21 columns.

Features Used:

Date

Hospital_ID

Hospital_Name

City

State

Disease_Type

Season

Patient_Count

ICU_Required

Beds_Required

Available_Beds

Oxygen_Cylinders_Required

Doctors_Required

Nurses_Required

Ventilators_Required
Medicine_1
Medicine_2
Medicine_3
Medicine_Units_Required
Occupancy_Rate
Emergency_Level

Machine Learning Methodology

The project uses supervised learning. Separate Random Forest models are trained for:

- Beds Required Prediction
- ICU Requirement Prediction
- Oxygen Requirement Prediction
- Doctor Requirement Prediction
- Medicine Units Prediction
- Emergency Level Classification

System Architecture

The frontend collects hospital details from users. JavaScript sends the data to the FastAPI backend through REST API endpoints. The backend preprocesses data and loads trained machine learning models to generate predictions.

Input and Output

Inputs:

- Patient Count
- Disease Type
- Season
- Occupancy Rate
- Available Beds

Outputs:

- Beds Required
- ICU Required
- Oxygen Required
- Doctors Required
- Medicine Units

- Emergency Level

Deployment

Frontend deployed using Vercel.
Backend and ML APIs deployed using Render.
Models stored using Joblib serialization.

Conclusion

The project successfully demonstrates a complete machine learning workflow integrated with a modern full-stack web application for healthcare resource demand forecasting.

Dataset Preview

Date	Hospital_ID	Hospital_Name	City	State	Disease_Type
2024-01-01	1079	Deleon LLC Hospital	South Maureenmouth	Hawaii	COVID
2024-01-02	1050	Bush-Waters Hospital	Lake Alyssa	Delaware	Dengue
2024-01-03	1061	Matthews Inc Hospital	New Kevinview	Hawaii	Flu
2024-01-04	1006	Harris-Martinez Hospital	Longburgh	Georgia	Typhoid
2024-01-05	1046	Henson LLC Hospital	Lauraville	New Jersey	Flu

Project Screenshots



Prediction Results

Beds Required: -

ICU Required: -

Oxygen Required: -

Doctors Required: -

Medicine Units: -

Emergency Level: -

Recent Alerts

ICU occupancy is above 75% in General Ward

10:30 AM

Oxygen level is below 65%

09:45 AM

